

Match the *most appropriate* term to each concept. Not all terms are used. No term is used twice.
(2 points each)

compensation depth	gyre	nanoplankton	resource partitioning
compensation intensity	holoepipelagic	net plankton	sea floor spreading
competition	holonekton	PAR	subduction
competitive exclusion	holoplankton	phytoplankton	thermocline
conveyor	holotrophic	protozoans	zooplankton
exclusive niche	<i>K</i> -type life history	<i>r</i> -type life history	
fundamental niche	mixoplankton	realized niche	

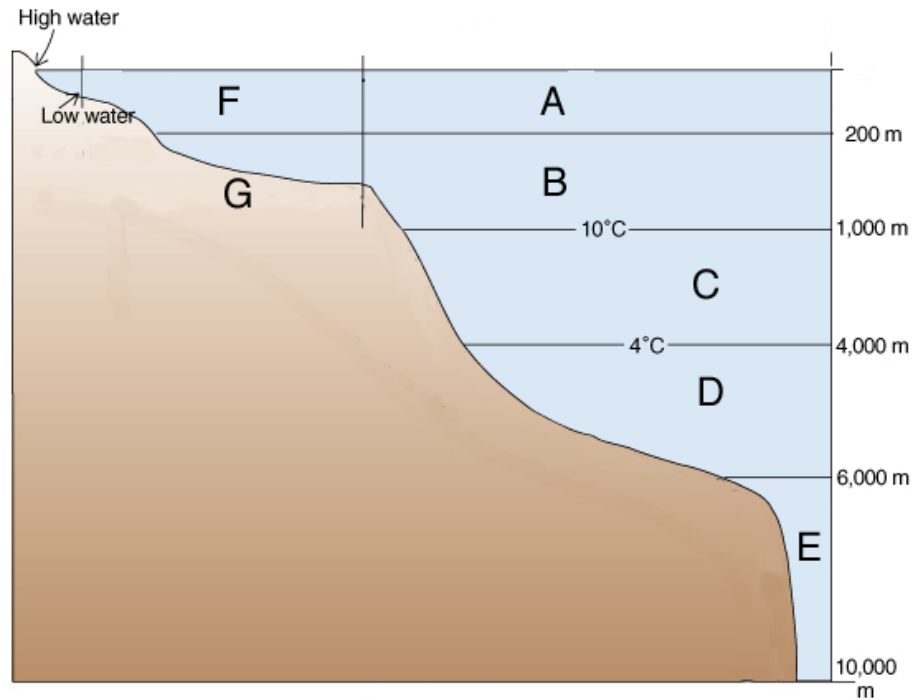
1. _____ Amount of light that can be used for photosynthesis.
2. _____ organisms grow fast, produce many offspring.
3. _____ Large circular surface current patterns.
4. _____ Non-swimming organisms less than 20 μm in size.
5. _____ Spend entire life free-swimming in surface waters.
6. _____ No two species can occupy the same niche.
7. _____ Transition zone for water temperature and density.
8. _____ Both heterotrophic and autotrophic.
9. _____ Creates new sea floor.
10. _____ Light levels sufficient for photosynthesis to equal respiration.

(2 pts) 11. What does PIC stand for?

(GSH) Answer the next two questions. Terms are not in the word bank. (2.5 points each.)

12. _____ Scientific name for class of diatoms.
13. _____ Scientific name for class of dinoflagellates.

Use correct names to fill in the blanks below for the zones identified by letters in the figure below. (1 point each).



14. _____ A, based on depth.
15. _____ B, based on light.
16. _____ C, based on depth.
17. _____ C, D, and E combined, based on light.
18. _____ F, open water out to shelf break.
19. _____ G, benthic, from low water mark.
20. _____ A and F combined, across ocean from shore to shore.
21. _____ Benthic between low and high water. Not lettered.

Extra credit: Proper names for D and E. (1 point each.)

D. _____ E. _____

Read each question carefully and completely. Be sure your answer addresses the question being asked. Use complete sentences.

- (5 pts) 22. Briefly explain two ways by which NPP is exported away from hydrothermal vents to the rest of the deep ocean.
- (5 pts) 23. What effect does lecithotrophic larval dispersal tend to have on the geographic range of a coastal invertebrate species compared to direct developing larvae? Briefly explain why.
- (10 pts) 24. Explain the one factor that both Cronin and Raymo (1997) and Rex et al. (2000) concluded was the major influence on benthic deep-sea diversity.
- (10 pts) 25. Tropical epipelagic phytoplankton “leak” as much as 25% of their DOM, so DOM should increase but it instead remains constant. Name and explain the process that keeps DOM in the epipelagic and returns it to the tropical food web. You may use an illustration but you must still explain it.

(10 pts) 26. Name and briefly explain the stages of community development associated with whale fall.

(10 pts) 27. Dr. Trieste d'Alvin compared benthic communities at 1000 m and 3000 m depth at several locations in the Atlantic Ocean. She discovered that benthic diversity was greater at 1000 m than at 3000 m. Briefly describe how spatial heterogeneity and sediment size might explain her results.

- (20 pts) 28. Summarize the classic model of primary productivity in the North Atlantic Ocean in the spring and early summer. Your summary should name the major phytoplankton and zooplankton organisms and describe their roles in overall net primary production. Your summary should also describe the seasonal availability of nutrients and light to explain the timing of peak productivity. You may include an illustration to aid your answer. **Grad and Honors students must also explain why this model does not work in the northern Pacific Ocean. (25 points for GSH.)**

- (20 pts) 29. **(GSH)** Explain what is the great ocean conveyor, how it works, and its importance to deep-sea communities. Then, predict what you think might happen to the conveyor and deep-sea communities if global climate change causes sea surface temperatures to increase by 3–4°C by the end of the 21st century. In addition, consider especially that warming of the atmosphere has been greatest at high latitudes, close to glacial regions. Justify your prediction. (20 pts)